

Qixuan Li

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RESEARCH INTERESTS

Robot planning and optimization, particularly **zeroth-order and distributional methods** for trajectory and policy optimization. Related interests include task and motion planning, contact-rich manipulation, and mobile manipulation.

EDUCATION

- **Tsinghua University – Shenzhen International Graduate School** Sep. 2023 – Jun. 2027 (Expected)
M.S. Candidate in Data Science and Information Technology Shenzhen, China
- **Xi'an Jiaotong University** Sep. 2019 – Jun. 2023
B.Eng. in Automation, Qian Xuesen Honors Program; additional undergraduate study in Economics Xi'an, China

SELECTED PUBLICATIONS AND MANUSCRIPTS

1. **Qixuan Li***, Chen Le*, Dongyue Huang, Jincheng Yu, and Xinlei Chen. "COMPASS: Confined-space Manipulation Planning with Active Sensing Strategy." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
2. **Qixuan Li***, Chen Le*, Jincheng Yu, and Xinlei Chen. "HCLM: A Hierarchical Framework for Cooperative Loco-Manipulation with Dual Quadrupeds." Under review at *IEEE Robotics and Automation Letters (RA-L)*.
3. **Qixuan Li**, Yanhong Zhao, and Jincheng Yu. "Experience-Guided Initiation Search for Learned Skills in Skill Composition." Under review at *IEEE International Conference on Robotics and Automation (ICRA)*, 2027.
4. Zhuozhu Jian*, **Qixuan Li***, et al. "Fault-tolerant cross-modal pose realignment for heterogeneous air-ground localization." Minor revision under review at *Tsinghua Science and Technology*.

RESEARCH EXPERIENCE

- **Confined-Space Manipulation Planning with Active Sensing (COMPASS)** 2024 – 2025
Tsinghua University [\[Code\]](#) [\[Paper\]](#) [\[Website\]](#)
 - Developed a planning framework for **robot manipulation under partial observability and tight kinematic constraints**.
 - Designed **MUE-RRT** to select exploration viewpoints by considering information gain, manipulability, motion cost, and task-specific guidance.
 - Integrated active exploration, motion planning, and **constrained grasp-pose optimization** into a closed-loop manipulation pipeline.
 - Validated the planning system in simulation and on a physical Franka manipulator, improving task success by **24.25%** over the strongest exploration baseline.
 - Led planning-algorithm development, system integration, and real-robot experiments.
- **Distributional Zeroth-Order Optimization for Constrained Robot Planning** 2026 – Present
Independent Research
 - Developing a **unified probability-space perspective** on population-based optimization through particle selection, transport, and mutation.
 - Designing **coupled optimality-feasibility paths** using Gibbs/Fisher-Rao tempering and constraint homotopy for constrained zeroth-order optimization.
 - Implementing a particle-based solver with reweighting, resampling, mutation, and constraint retraction, using **objective values and explicit constraint Jacobians**.
 - Benchmarked CEM, MPPI, CMA-ES, and SMC-based methods, analyzing **mode discovery, local refinement, and final solution quality** under limited budgets.
 - Built a **42D robot trajectory optimization** testbed in robosuite/MuJoCo, with dense feasibility validation and batched geometric constraint evaluation.
- **Cooperative Loco-Manipulation with Dual Quadrupeds (HCLM)** 2025 – 2026
Tsinghua University [\[Paper\]](#) [\[Video\]](#)
 - Developed a **hierarchical loco-manipulation framework** for dual quadruped manipulators, combining high-level task-space coordination with whole-body execution.
 - Integrated imitation-learning-based task-space motion generation with kinematic MPC, RL locomotion, QP-based arm control, collision avoidance, and closed-chain force regulation.
 - Built simulation and real-robot systems using **dual Unitree Go2-ARX5** platforms. Validated carrying, packing, and handover in simulation, and dynamic handover on physical robots.
 - Led RL locomotion training, system integration, and real-robot deployment; contributed to system design.
- **Experience-Guided Initiation Search for Learned Skill Composition** 2026
Tsinghua University [\[Paper\]](#)
 - **Addressed reliable initiation selection** for frozen learned skills under limited interaction budgets, where individual-skill success does not guarantee **full-sequence success**.

- Developed **EVIS**, a propose–validate–recover framework combining historical execution experience with behavioral validation and recovery in new environments.
 - Evaluated EVIS with VLA policies in RoboCasa365. Led algorithm design, experimental evaluation, and analysis.
- **Fault-Tolerant Cross-Modal Pose Realignment for Heterogeneous Air–Ground Localization** 2023 – 2024
Tsinghua University
- Addressed cross-modal pose alignment failures caused by **modality differences and large pose errors** in heterogeneous air–ground robot systems.
 - Formulated geometric visual–LiDAR associations using point-to-plane constraints for robust pose alignment.
 - Combined event-triggered **Particle Swarm Optimization (PSO)** for global coarse alignment with tightly coupled visual–inertial–LiDAR nonlinear optimization for refinement.
 - Contributed to geometric optimization formulation and implementation, simulation, and hardware validation.

INDUSTRY AND ENGINEERING EXPERIENCE

- **Meituan – Autonomous Systems / UAV Division** Nov. 2024 – Apr. 2025
Algorithm Intern – Multi-Robot Motion Planning Shenzhen, China
 - Adapted an EGO-Swarm-based **trajectory optimization framework** for coordinated motion planning of multiple autonomous ground vehicles.
 - Implemented trajectory representations, collision-avoidance costs, and inter-robot coordination mechanisms; evaluated the planning pipeline in multi-agent simulations.
- **Wearable Exoskeleton Startup Project** 2026 – Present
Founding Technical Team Member Shenzhen, China
 - Developed low-latency models for short-horizon knee-torque prediction and gait-event estimation from multi-sensor human-motion data.
 - Investigated **data-efficient personalization** through low-dimensional adaptation and active/Bayesian optimization with limited per-user calibration data.
 - Contributed to early-stage **product development**, spanning sensor-data collection, algorithm evaluation, and embedded deployment considerations.
- **RoboMaster Robotics Team, Xi'an Jiaotong University** 2020 – 2022
Control / Electrical Systems Team; later Control Team Lead Xi'an, China
 - Developed robot electrical and control systems, including an **LQR-based balance controller** for a bipedal wheeled robot.
 - Gained **full-stack robotics experience** in sensing, embedded control, actuation, and system integration and debugging.
 - Led the control subgroup, coordinating controller development and cross-subsystem integration.

TECHNICAL SKILLS

Robot Planning & Formulation: Trajectory optimization, constrained motion planning, active exploration, geometric and kinematic constraint modeling.

Optimization Theory & Methods: Zeroth-order and distributional optimization, probability-space gradient flows, Gibbs/Fisher–Rao dynamics, constraint homotopy, particle-based optimization.

Optimization Algorithms & Solvers: CEM, MPPI, CMA-ES, SMC, PSO, Gaussian-process-based search; IGO and SVGD.

Robot Control: LQR, PID, impedance control, kinematic MPC, QP-based control, RL-based whole-body control.

Robot Learning: Reinforcement-learning locomotion, imitation learning, diffusion policies, VLA policy fine-tuning and deployment.

Programming & Software: Python, C/C++, PyTorch, ROS/ROS 2, Linux.

Simulation & Platforms: Isaac Gym, Isaac Sim, MuJoCo; Franka manipulators, Unitree Go2–ARX5, UAV/UGV platforms.

SELECTED HONORS AND AWARDS

First Prize , RoboMaster 2022 National Competition	2022
First Prize , RoboMaster 2022 Central Regional Competition	2022
Provincial First Prize , China Undergraduate Mathematical Contest in Modeling	2020
Provincial First Prize , National Undergraduate Mathematics Competition	2020